

Practical: Word Count using Hadoop MapReduce

This document explains step-by-step instructions to execute Practical No. 1 of the DSBD Lab Group B Section, which involves creating a simple Word Count application using the Hadoop MapReduce framework on a local standalone setup.

1. Install Hadoop and set up the Hadoop Distributed File System (HDFS) on your PC.
2. Open Command Prompt (CMD) and check Java and Hadoop versions using commands:
'java -version' and 'hadoop version'.
3. Run CMD as Administrator and start Hadoop services using: **'start-all.cmd'**.
4. Check all Hadoop services with the command: **'jps'**. **Ensure DataNode, NameNode, ResourceManager, and NodeManager are running.**
5. Open a browser and visit **'http://localhost:9870'** to verify HDFS is working. Also, check YARN on **'http://localhost:8088'**.
6. Create a directory in HDFS using: **'hadoop fs -mkdir /input'**.
7. Create a text file (e.g., page1.txt) with sample words in your local system.
8. Upload the text file to HDFS using: **'hadoop fs -put [local_path] /input'**.
9. Verify the file is uploaded using: **'hadoop fs -ls /input'**.
10. Open Eclipse IDE, create a new Java Project named **'MapReduceWordCount'**.
11. Set up the project with Java SE 1.8 and create a package **'com.mapreduce.wordcount'**.
12. Add all necessary Hadoop JAR files from the installation directory to the project build path.

13. Create a new Java class named 'WordCount' and paste the MapReduce code into it.
14. Ensure the package and class names match, and there are no errors in the code.
15. Export the project as a JAR file (e.g., wordcount-mapreduce.jar).
16. Run the JAR file using the command: **'hadoop jar [jar_path]**
com.mapreduce.wordcount.WordCount /input/page1.txt /output'.
17. Wait for the job to complete and check output **in HDFS '/output'** directory.
18. View the output using: **'hadoop fs -cat /output/*'**.
19. (Optional) Download output from HDFS to local file system using: **'hadoop fs -get**
/output [local_path]'.